

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages	Python, TypeScript, JavaScript, Java, SQL, C++, Bash
LLM & Agentic Systems	LLM agents, tool use / function calling, reasoning & planning, prompt engineering, context management, agent memory / persistent state, guardrails, provider routing/fallback, OpenAI / Groq APIs, Ollama
Retrieval / RAG	RAG architecture, chunking, embeddings (sentence-transformers), semantic retrieval, RRF, cross-encoder reranking, MMR, vector search (FAISS), llama-index, spaCy
Backend & Cloud	FastAPI, Node.js, Express, REST APIs, async services, React, WebSocket / streaming, AWS (S3, KMS, SQS), GCP, Azure, Docker, Git, GitHub Actions
ML / Deep Learning	PyTorch, JAX, HuggingFace Transformers, NumPy, normalizing flows, diffusion, autodiff

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by AI team to label data for production models; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**, accelerating iteration on downstream NLP/clinical models.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and remediated **20+ VAPT findings**.
- Selected for a measured **GitHub Copilot** productivity pilot in an **Agile/Scrum** team, routing tech-debt PRs, code migrations, and security-finding triage through AI-assisted workflows and maintaining a versioned conventions file to keep AI-generated code aligned with team patterns.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Integrated **LLMs** into a chat product to generate context-aware replies from conversation history and user input; iterated on prompt design and response-quality heuristics for a real customer-facing experience while building 20+ Node.js + Fastify REST APIs over PostgreSQL.

Projects

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, Brave Search, SQLite

- Built a multi-stage **LLM agent** that plans typed retrieval, calls search and extraction tools, and verifies its own outputs, converting open-ended queries into structured tables with **cell-level provenance** (value, source, confidence) for every result.
- Engineered a well-tested **FastAPI** service with **150+ automated tests**, provider routing/fallback across OpenAI and Groq as a reliability guardrail, deterministic extraction before LLM fallback, and a reviewer-facing trust UI.

KG2RAG-Enhanced - Knowledge-Grounded RAG for Multi-Hop QA

github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama + LLaMA-3, sentence-transformers, llama-index, spaCy | UMass CS 685 NLP

- Led the **multi-view retrieval** component of a team **RAG** pipeline grounded on a knowledge base, query-side sub-question decomposition, per-view embedding retrieval, RRF fusion, cross-encoder reranking, and MMR, built on **llama-index** and sentence-transformer embeddings.
- Lifted supporting-fact recall **+2.68%** (54.53 to 57.21) over the KG²RAG baseline on a **4,905-question** evaluation with token and latency logging and 95% confidence intervals.

cf-ai-research-scout - Conversational AI Research Agent

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Workers AI, AIChatAgent, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built a conversational research **agent** that fetches web sources via tool use, generates multi-step digests, and **streams** responses to the client over **WebSocket** in real time, using embeddings for retrieval.
- Implemented **persistent agent memory** and durable multi-step workflows (Durable Objects + WorkflowEntrypoint + D1) so conversation state and long-running tasks survive process restarts and tool failures.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.